

## NANCHENG, China

WMO: 587150

Lat: 27.5767N

Lon: 116.6385E

Elev: 269

StdP: 14.55

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 30.8       | 32.5 | 19.0                          | 14.4 | 41.3 | 23.3 | 17.8 | 41.4 | 17.4                  | 44.3 | 15.7 | 46.5 | 5.2                   | 340  | 0.382 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.5                   | 96.8            | 78.8 | 95.1 | 78.7 | 93.3 | 78.4 | 81.5                | 90.8 | 80.7 | 90.0 | 79.9 | 88.8 | 6.3                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 79.2                            | 152.9 | 85.9 | 78.3 | 148.6 | 85.0 | 77.6 | 145.1 | 84.3 | 45.6          | 91.0 | 44.7 | 90.3 | 43.9 | 89.0 | 85.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 16.1              | 14.0 | 12.3 | DB                         | 26.9 | 99.4               | 2.6 | 2.1                                                | 25.0 | 100.9      | 23.4 | 102.1      | 22.0 | 103.3      | 20.0 | 104.9 |
|                   |      |      | WB                         | 25.5 | 83.2               | 2.8 | 1.1                                                | 23.5 | 84.0       | 21.9 | 84.6       | 20.3 | 85.3       | 18.3 | 86.1  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 66.2        | 44.2  | 48.7  | 56.1  | 66.7  | 74.5  | 79.7  | 85.5  | 84.4  | 78.3  | 69.2  | 58.5  | 47.7  | (6)  |
|                                                                        | DBStd       | 15.51       | 7.07  | 8.87  | 8.17  | 7.27  | 4.97  | 4.90  | 3.91  | 4.09  | 5.74  | 5.81  | 7.07  | 6.18  | (7)  |
|                                                                        | HDD50       | 495         | 211   | 121   | 32    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 13    | 117   | (8)  |
|                                                                        | HDD65       | 2251        | 646   | 461   | 297   | 65    | 2     | 0     | 0     | 0     | 0     | 26    | 217   | 537   | (9)  |
|                                                                        | CDD50       | 6409        | 30    | 86    | 220   | 502   | 760   | 890   | 1102  | 1065  | 848   | 594   | 267   | 45    | (10) |
|                                                                        | CDD65       | 2692        | 0     | 6     | 20    | 116   | 297   | 440   | 637   | 600   | 399   | 155   | 21    | 1     | (11) |
|                                                                        | CDH74       | 26749       | 3     | 36    | 126   | 696   | 2035  | 4112  | 8050  | 7107  | 3611  | 862   | 108   | 3     | (12) |
|                                                                        | CDH80       | 11292       | 0     | 3     | 19    | 168   | 627   | 1535  | 4026  | 3276  | 1399  | 222   | 17    | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 6.1         | 6.1   | 6.4   | 6.3   | 6.5   | 6.0   | 6.2   | 6.9   | 5.9   | 5.6   | 5.7   | 5.9   | 6.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 64.9        | 2.6   | 4.3   | 7.1   | 9.7   | 10.6  | 11.5  | 4.6   | 4.3   | 3.5   | 2.4   | 2.5   | 1.8   | (15) |
|                                                                        | PrecMax     | 84.2        | 6.3   | 8.9   | 11.1  | 21.1  | 18.0  | 21.1  | 12.8  | 13.0  | 8.2   | 5.9   | 7.1   | 4.5   | (16) |
|                                                                        | PrecMin     | 35.6        | 0.2   | 0.4   | 1.6   | 3.5   | 2.4   | 3.7   | 0.2   | 0.4   | 0.2   | 0.0   | 0.0   | 0.0   | (17) |
|                                                                        | PrecStd     | 11.9        | 1.6   | 2.1   | 2.7   | 3.7   | 4.3   | 4.9   | 3.3   | 3.1   | 2.4   | 1.8   | 2.0   | 1.4   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 70.7  | 78.6  | 82.8  | 88.6  | 92.8  | 95.4  | 99.6  | 99.6  | 96.3  | 90.7  | 82.3  | 71.1  | (19) |
|                                                                        |             | MCWB        | 60.5  | 64.7  | 67.4  | 71.8  | 74.6  | 78.9  | 78.9  | 78.2  | 78.5  | 74.6  | 67.7  | 58.6  | (20) |
|                                                                        | 2%          | DB          | 64.8  | 73.4  | 77.4  | 84.5  | 89.4  | 92.9  | 97.1  | 96.7  | 93.6  | 85.7  | 77.3  | 64.9  | (21) |
|                                                                        |             | MCWB        | 55.8  | 61.7  | 64.8  | 70.3  | 73.6  | 77.9  | 78.9  | 78.8  | 77.6  | 71.6  | 64.1  | 55.6  | (22) |
|                                                                        | 5%          | DB          | 59.4  | 67.5  | 73.3  | 81.3  | 86.4  | 90.6  | 95.4  | 94.7  | 90.7  | 82.0  | 73.0  | 61.0  | (23) |
|                                                                        |             | MCWB        | 51.8  | 58.8  | 62.7  | 69.1  | 72.8  | 77.8  | 78.7  | 79.1  | 77.3  | 68.9  | 62.6  | 53.5  | (24) |
|                                                                        | 10%         | DB          | 54.6  | 62.3  | 68.7  | 78.0  | 83.5  | 88.2  | 93.6  | 92.6  | 87.8  | 78.8  | 69.2  | 57.4  | (25) |
|                                                                        |             | MCWB        | 49.1  | 55.5  | 60.9  | 68.0  | 72.8  | 77.4  | 78.5  | 78.9  | 76.5  | 67.6  | 61.6  | 51.0  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 63.0  | 66.5  | 70.2  | 74.8  | 78.7  | 81.5  | 82.7  | 82.6  | 81.6  | 76.5  | 71.0  | 63.3  | (27) |
|                                                                        |             | MCDB        | 69.8  | 74.6  | 78.8  | 83.6  | 86.1  | 89.8  | 92.5  | 91.7  | 90.2  | 85.6  | 77.3  | 67.8  | (28) |
|                                                                        | 2%          | WB          | 57.0  | 62.9  | 67.5  | 72.9  | 77.3  | 80.1  | 81.5  | 81.5  | 79.9  | 74.3  | 67.7  | 58.4  | (29) |
|                                                                        |             | MCDB        | 62.4  | 71.7  | 74.5  | 81.1  | 84.3  | 88.4  | 91.2  | 90.8  | 88.4  | 81.3  | 72.5  | 62.1  | (30) |
|                                                                        | 5%          | WB          | 53.0  | 59.8  | 64.9  | 71.5  | 76.0  | 79.1  | 80.7  | 80.6  | 78.8  | 72.4  | 65.6  | 54.7  | (31) |
|                                                                        |             | MCDB        | 57.5  | 66.5  | 71.4  | 78.6  | 82.4  | 87.1  | 90.1  | 90.0  | 87.0  | 78.5  | 70.2  | 59.1  | (32) |
|                                                                        | 10%         | WB          | 49.3  | 56.0  | 61.9  | 69.9  | 74.6  | 78.3  | 79.9  | 79.9  | 77.9  | 70.5  | 62.8  | 51.9  | (33) |
|                                                                        |             | MCDB        | 53.7  | 60.9  | 67.2  | 75.5  | 80.5  | 85.6  | 89.2  | 88.6  | 85.3  | 76.0  | 67.7  | 55.9  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.5  | 11.6  | 12.5  | 13.0  | 12.8  | 11.1  | 13.5  | 12.9  | 12.9  | 13.9  | 13.5  | 12.7  | (35) |
|                                                                        |             | MCDBR       | 21.1  | 21.6  | 21.5  | 19.2  | 18.9  | 15.3  | 16.4  | 16.7  | 16.9  | 19.5  | 20.8  | 21.1  | (36) |
|                                                                        | 5% WB       | MCWBR       | 12.0  | 11.6  | 9.8   | 7.7   | 6.4   | 4.7   | 4.2   | 4.1   | 4.5   | 6.5   | 8.8   | 11.3  | (37) |
|                                                                        |             | MCWBR       | 16.8  | 19.2  | 17.6  | 15.1  | 13.3  | 13.1  | 14.6  | 14.7  | 14.5  | 14.4  | 13.7  | 15.2  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.655 | 0.699 | 0.770 | 0.743 | 0.658 | 0.631 | 0.584 | 0.630 | 0.642 | 0.663 | 0.612 | 0.600 | (40) |
|                                                                        |             | taud        | 1.573 | 1.529 | 1.449 | 1.535 | 1.732 | 1.804 | 1.936 | 1.801 | 1.744 | 1.640 | 1.689 | 1.676 | (41) |
|                                                                        | Ebn at Noon |             | 183   | 190   | 187   | 200   | 218   | 223   | 233   | 222   | 213   | 195   | 193   | 189   | (42) |
|                                                                        |             | Edh at Noon | 75    | 84    | 96    | 91    | 75    | 69    | 60    | 69    | 71    | 74    | 66    | 65    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 637   | 744   | 902   | 1103  | 1260  | 1281  | 1769  | 1540  | 1313  | 1111  | 832   | 739   | (44) |
|                                                                        | RadStd      |             | 147   | 187   | 115   | 160   | 172   | 167   | 151   | 138   | 120   | 189   | 146   | 164   | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | +0.83 | N/A     | N/A    | +1.13   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (10 neighbors) | +0.63 | N/A     | N/A    | +0.97   | N/A   | N/A   | N/A         | N/A   | +247  | +166  |

Nomenclature: See separate page